Mend-Amar Badral

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Practical, systems-focused builder with hands-on Linux experience, driven by problem-solving and a preference for simple, working solutions over complexity.

Education

Budapest University of Technology and Economics (BME) — Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering — Budapest, Hungary
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) — Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics — Ulaanbaatar, Mongolia
GPA: 3.2/4.0 **(rank 3/34)**
Supervisor: Galtbayar Artbazar
Thesis: Evaluating land degradation using image processing methods

Experience

Center of Mathematics Application Research Lab — Oct, 2023 – June, 2025
Undergraduate Lab Lead & Research Intern

- Selected as undergraduate Lab Lead; participated in weekly seminars and collaborative discussions with faculty members on machine learning, statistical modeling, and applied image analysis.
- Developed a custom vendor-specific **Python pipeline** for processing multispectral drone imagery and sensor metadata to generate vegetation and land degradation maps.
- Worked extensively with multispectral image data, vegetation indices (NDVI, MSAVI), raster data processing, and geospatial analysis using Python tools including NumPy, OpenCV, Rasterio, and GDAL.
- Implemented feature engineering and threshold optimization approaches for interpretable land classification and environmental analysis from UAV imagery.

Projects

Data Integration for 3D Gaussian Splatting

- Implemented a custom JSON parser to extract sensor data obtained by aiSim, an autonomous driving simulator by aiMotive, into a nerfstudio (3D reconstruction software) format. Performed LiDAR point cloud sensor fusion with the camera images to obtain color information.

Deep Learning Foundations and Concepts Book Examples Reproduction

- Implemented polynomial regression from Bishop & Bishop's *Deep Learning Foundations and Concepts*, and reproduced key theoretical results in interactive Julia notebooks to empirically analyze **overfitting, underfitting, and the bias–variance tradeoff**.

Awards

Stipendium Hungaricum Scholarship — Sep, 2025

Skills

Languages:	English (C1 IELTS 8.0)
Programming:	Python, Bash, JavaScript, Julia, C/C++, R
Tools:	Git, pandas, sklearn, rasterio, PyTorch, GDAL, OpenCV, nerfstudio, QGIS, ArcGIS